ESEQUIEL LINARES

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EDUCATION

Purdue University West Lafayette, IN
Bachelor of Science in Computer Science, Machine Intelligence Concentration *May 2028*

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, Linear Algebra, Multivariate Calculus, Discrete Math, Intro to AI, Systems Programming

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C, JavaScript, TypeScript, SQL, Swift, Bash
Frameworks & Libraries: FastAPI, Flask, Spring Boot, React, Next.js, Node.js, NumPy, Pandas, Mediapipe, JUnit, pytest
AI/ML: PyTorch, Hugging Face, llama.cpp, faster-whisper, LLM APIs (Claude, Gemini), RAG
Tools & Platforms: Git, Docker, Databricks, Delta Lake, PySpark, Jenkins, Bitbucket, AWS (EC2), GCP, Supabase, Modal, Linux/Unix, PostgreSQL

EXPERIENCE

Software Engineering Intern June 2026 – August 2026
JPMorgan Chase & Co. *Chicago, IL*

- Streamlined end-to-end processing for a 100-document batch from ˜2 hours to 11 minutes by profiling the bottleneck, parallelizing independent documents across 4–8 workers, and batching per-record Delta writes into 300-record merges.
- Designed and built the ingestion layer of Hermes, an internal document processing platform converting invoices and contracts (PDF, PNG) into schema-validated Delta tables, so new file formats and parsers drop in without pipeline code changes.
- Implemented validation gates for the LLM mapping job, layering deterministic field checks with an LLM verification pass and routing malformed records to a Dead Letter Queue, with JSON schemas and guardrail prompts loaded from Delta tables rather than hardcoded.
- Presented the platform to senior firm leadership, securing handoff to a partner team for production adoption.
- Mentored two high school interns building the RapidOCR extraction path.

Co-Founder & Lead Engineer January 2026 – Present
Estovam Tech (estovam.tech) *West Lafayette, IN*

- Architected split compute for an autonomous voice-and-vision robot, pairing a Raspberry Pi 5 orchestrator with a Qualcomm QCS6490 AI coprocessor that runs offloaded speech-to-text and LLM inference.
- Benchmarked on-device LLMs and selected a quantized 3B model (5.2 tok/s, 8K context) over an 8B variant, preserving agentic tool use.
- Diagnosed on-device TTS on the Adreno 643 GPU running at 0.04x realtime, sharing memory with the CPU and competing with the on-device LLM; traded it for sentence-streaming cloud TTS (ElevenLabs), cutting time-to-first-audio from ˜60s to under 2s.

Undergraduate Researcher – Multimodal Generative AI September 2025 – May 2026
Purdue University *West Lafayette, IN*

- Extended PCMDM, a past-conditioned diffusion model for coherent long-form 3D human motion generation, training a variant on a 10K+ annotation duet dance dataset that improved R-Precision Top-3 by 60% relative to the provided baseline.
- Built PyTorch training and evaluation pipelines for Text-to-Duet and Dance Accompaniment, implementing an R-Precision harness that ranks ground-truth captions against mismatched candidates in a shared text-motion embedding space.

PROJECTS

Price is Right Glasses | *FastAPI, Claude/Gemini Vision, Supabase, SwiftUI, ElevenLabs*

- Built a Meta Ray-Ban glasses assistant that identifies items by vision, checks live eBay and Discogs prices, and speaks a verdict.
- Engineered a provider-agnostic FastAPI vision layer with swappable Claude and Gemini models behind one prompt and response schema.

machine(learn) | *Modal, Next.js, TypeScript, Supabase, Pydantic*

- Orchestrated an autonomous ML agent swarm on Modal GPUs with a four-phase async pipeline and typed Pydantic agent handoffs.
- Streamed live run state to a Next.js dashboard over Supabase realtime, with authentication and per-run dataset uploads.

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Software Engineering Fellow September 2026 – Present
Google x Basta Code2Career *Remote*

- Selected for a competitive 10-week technical interview program run by Basta in partnership with Google.

Career Prep Fellow January 2026 – Present
Management Leadership for Tomorrow

- Selected for an 18-month program accelerating emerging technical leaders through coaching, technical assessments, and project work.

Backend Developer – Code for Good Hackathon November 2025
JPMorgan Chase & Co. *Columbus, OH*

- Led the backend architecture, building 10+ Flask REST endpoints (SQLAlchemy ORM) and an automated CSV ingestion pipeline that turned raw data into structured records for real-time dashboards.
- Deployed the backend to AWS EC2, contributing to a top-3 finish within 24 hours.